# Practice Questions Sheet 2

This tutorial introduces **contingency tables and ways to analyse and visualise categorical data**. Please **attempt all questions** before your tutorial You should work through the questions in a new R script.
For questions asking you to interpret a result, **aim for one or two clear sentences**. You should use the context of the insurance data when explaining your answers.

Student name: _____

Student number: _____

**Please complete this assignment before the tutorial on the week of 2nd November**

# Insurance data

We will use a dataset containing information of 150 customers of an insurance company. Within this dataset, these people were asked asked about their satisfaction with their motor insurance. Each customer was classified according to:

- `satisfaction`: satisfied or unsatisfied with their insurance cover
- `vehicle`: What type of vehicle they own (petrol car, diesel car, or motorbike)

The dataset is available on the RStudio server in the ST201 data folder.
You can load it by opening a new R script and running the following code:

```
# Load the insurance dataset
insurance <- readRDS("Sharedfiles/ST201/data/insurance.rds")
```

Once you have loaded the data, you can look at the first few observations using:

```
# Examine the first 6 rows of the insurance dataset
head(insurance)
```

## Exercise 1

- How many **observations** are in the dataset
- What are the two **categorical variables**?
- **How many categories** does each variable contain?
- What is the **observational unit** in this dataset?

# Contingency tables

A contingency table allows us to examine the relationship between two categorical variables.

Table 1: Contingency table of the insurance dataset

| Satisfaction | Diesel car | Motorbike | Petrol car |
|---|---|---|---|
| Satisfied | 29 | 12 | 33 |
| Unsatisfied | 31 | 20 | 25 |

## Exercise 2

- Create a contingency table (similar to the above) of the insurance data.
- Use the `addmargins()` function to add the marginal frequencies to the table.
- How many customers
  - own petrol cars?
  - own diesel cars?
  - own motorbikes?
  - are satisfied with their insurance?
  - are unsatisfied with their insurance?
- Which vehicle type has the largest number of customers in the sample?

# Proportions

Frequencies can be useful. However, proportions can oftentimes make comparisons easier.

### Exercise 3

- Turn the contingency table you made in exercise 2 into a proportions table.
- Again, use the `addmargins()` function to add the marginal proportions to the table.
- What proportion of customers were satisfied with their insurance?
- What proportion of customers owned a petrol car?
- Explain the difference between the proportion in the cell
  - *"petrol car and satisfied"* and *"marginal proportion satisfied"*

# Conditional distributions

Suppose we want to compare satisfaction **within each vehicle type**.

$$P(\text{satisfaction} \mid \text{vehicle type})$$

### Exercise 4

- Update the proportion table from exercise 3 to now show the conditional proportions.
- What proportion of petrol-car customers were satisfied?
- What proportion of diesel-car customers were satisfied?
- What proportion of motorbike customers were satisfied?
- Which vehicle type has the highest proportion of satisfied customers?
- Based on these conditional proportions, describe the relationship between vehicle type and satisfaction.

## Visualising categorical data

A bar plot lets us compare the numbers of satisfied and unsatisfied customers across vehicle types.

```
# args.legend: Changes the legend position
# names.arg: Change the text on the x-axis
# ylim: Changes the y-axis limits

barplot(
  insurance_table,
  legend = TRUE,
  args.legend = list(x = "topleft"),
  names.arg = c("Diesel car", "Motorbike", "Petrol car"),
  ylim = c(0, 110)
)
```

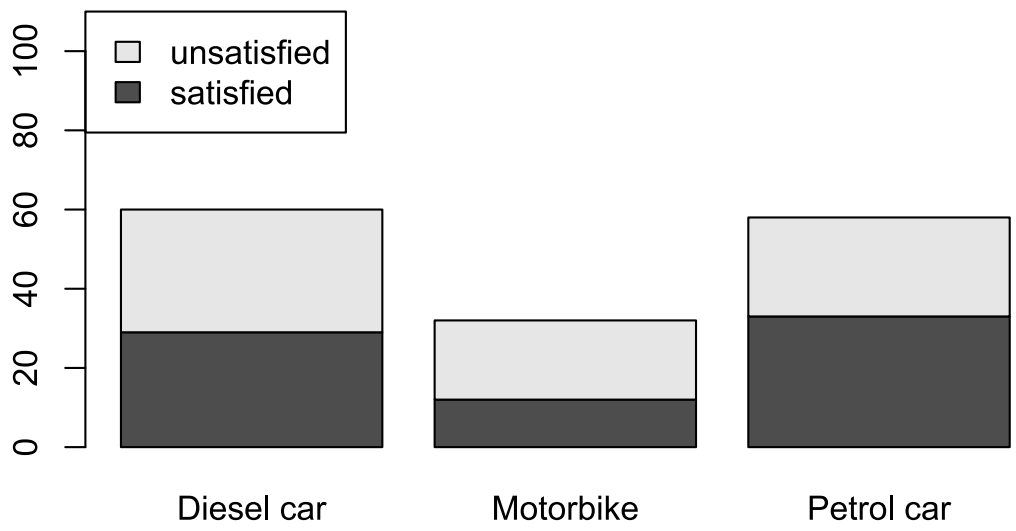



Figure 1: Example of a barplot showing the relationship between insurance satisfaction and vehicle type

### Exercise 5

- Instead of a stacked bar plot, modify the above code to create a side-by-side bar plot. You may also modify the `ylim` argument to improve the resulting figure
- Produce a stacked bar plot in which each bar represents 100% of the customers in that vehicle group.

# Odds and odds ratios

The odds of an outcome are defined as

$$\text{odds} = \frac{P(\text{outcome})}{1 - P(\text{outcome})}$$

Therefore, if 60% of a group are satisfied, then

$$\frac{0.6}{1 - 0.6} = 1.5$$

there are 1.5 satisfied customers for every 1 unsatisfied customer.

## Exercise 6

- Calculate the odds of satisfaction for
  - petrol-car customers
  - diesel-car customers
  - motorbike customers
- Calculate the odds ratio comparing petrol-car customers with diesel-car customers
- Interpret the odds ratio in the context of this study

# $\chi^2$ test

We now want to formally test whether vehicle type and satisfaction are independent. In order to do this, we will perform a Pearson's $\chi^2$ test.

### Exercise 7

- State what your null and alternative hypotheses are.
- Write R code to perform a $\chi^2$ test
  - What is the value of the chi-squared test statistic?
  - What are the degrees of freedom?
  - What is the p-value?
  - Using a 5% significance level, what decision would you make about the null hypothesis?
  - Write a conclusion for the test in the context of insurance satisfaction.